Crypto Strategy Backtester

Cost-aware research on fixed-rule cryptocurrency signals

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Sample	Assets	Initial capital
2024-01-01 to 2026-01-01	BTCUSDT, ETHUSDT, BNBUSDT	10,000 USDT

Executive summary

This study tests transparent mean-reversion and trend-following rules on 6-hour spot crypto data. The framework makes time alignment, target exposure, turnover, transaction costs, and failure conditions explicit. All executable targets use a one-bar signal lag.

At the baseline cost, full-sample trend following returns +188.99% but has a Sharpe ratio of 0.207 and a maximum drawdown of -82.10%. Mean reversion loses all capital. Most importantly, the fixed trend rule also loses almost all capital in the illustrative 2025 holdout.

The evidence does not support an investable-strategy claim. It supports a software and research conclusion: cost assumptions, chronological evaluation, and downside metrics can reverse the story told by a single profitable full-sample curve.

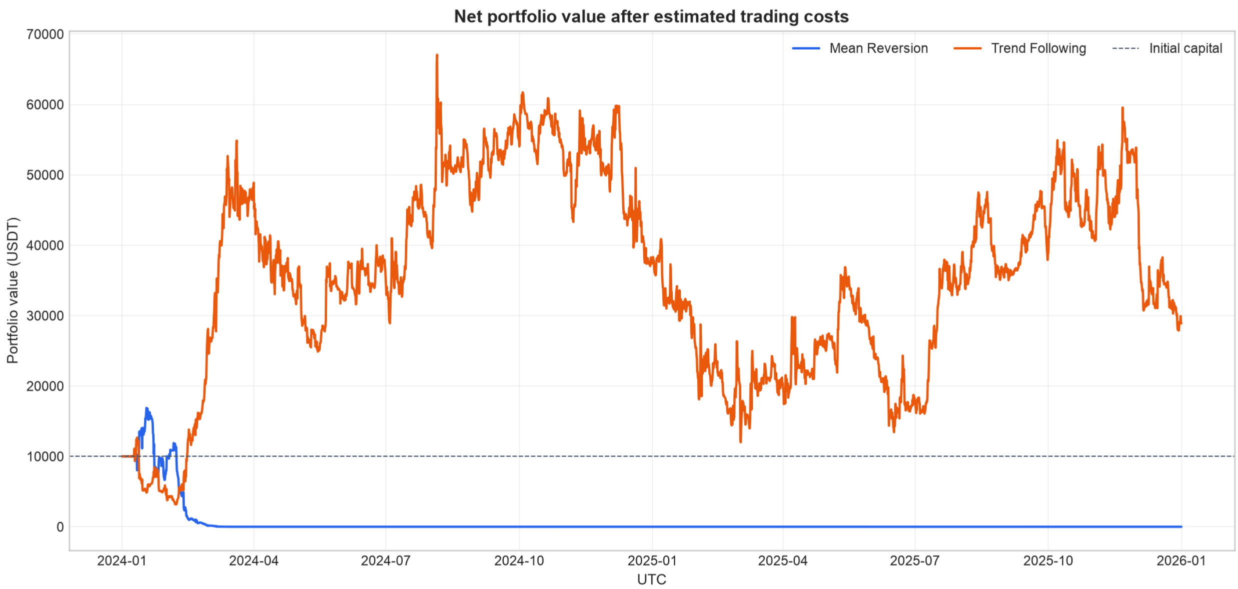

Figure 1. Full-sample net portfolio value after the baseline Roll-style trading cost.

1. Research design

Data

The sample contains 2,924 UTC bars per asset from Binance Public Data. The effective federal funds rate from FRED is converted to a 6-hour step rate and aligned using only same-date or earlier observations. The validation layer checks schema, symbol coverage, chronological ordering, duplicates, gaps, price positivity, and finite values.

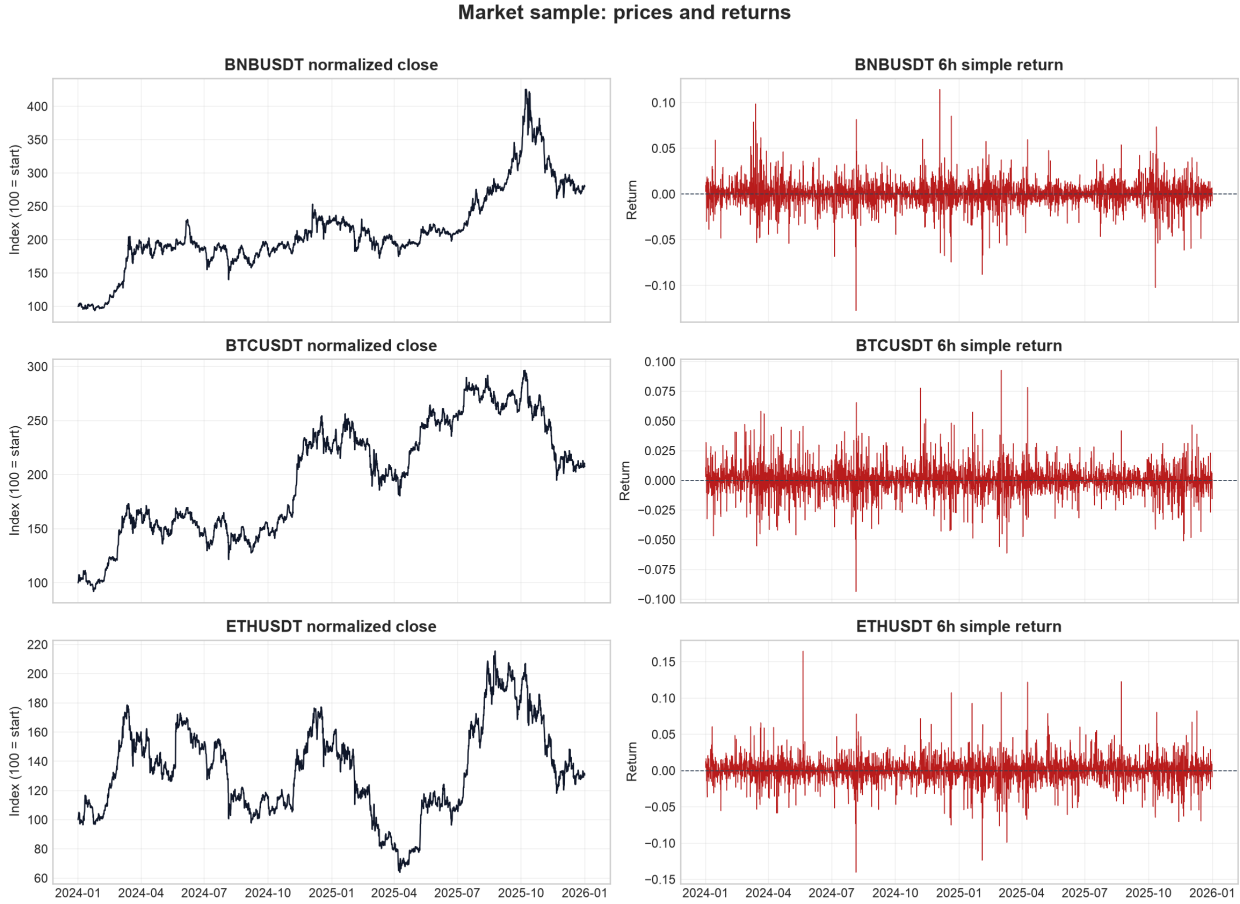

Figure 2. Normalized close prices and 6-hour simple returns for the three assets.

Signals and timing

Mean reversion uses a 40-bar rolling price z-score: long below -1 and short above +1. Trend following uses the sign of the 8-bar versus 32-bar moving-average difference. Signals observed at time t become executable targets at t+1. Target weights are normalized to a 100,000 USDT gross cap.

Cost and capital model

A per-asset Roll-style estimate is derived from lag-1 return covariance and averaged across assets. The baseline estimate is 0.1147% per unit traded, with 0.5x and 1.5x sensitivity cases. Executed gross exposure is limited to the smaller of 100,000 USDT and 10 times current equity. Realized portfolio value cannot become negative.

2. Full-sample results

Strategy	Net return	Sharpe	Sortino	Max drawdown	Turnover
Mean reversion	-100.00%	-0.680	-0.264	-100.00%	3,822,642
Trend following	+188.99%	0.207	0.277	-82.10%	25,688,850

Trend following finishes at 28,898.84 USDT, but modeled baseline cost totals 29,466.47 USDT. Its Sharpe ratio is low, and its 82.1% drawdown is too large to describe the rule as robust. Mean reversion cannot overcome turnover-linked cost and reaches the capital floor.

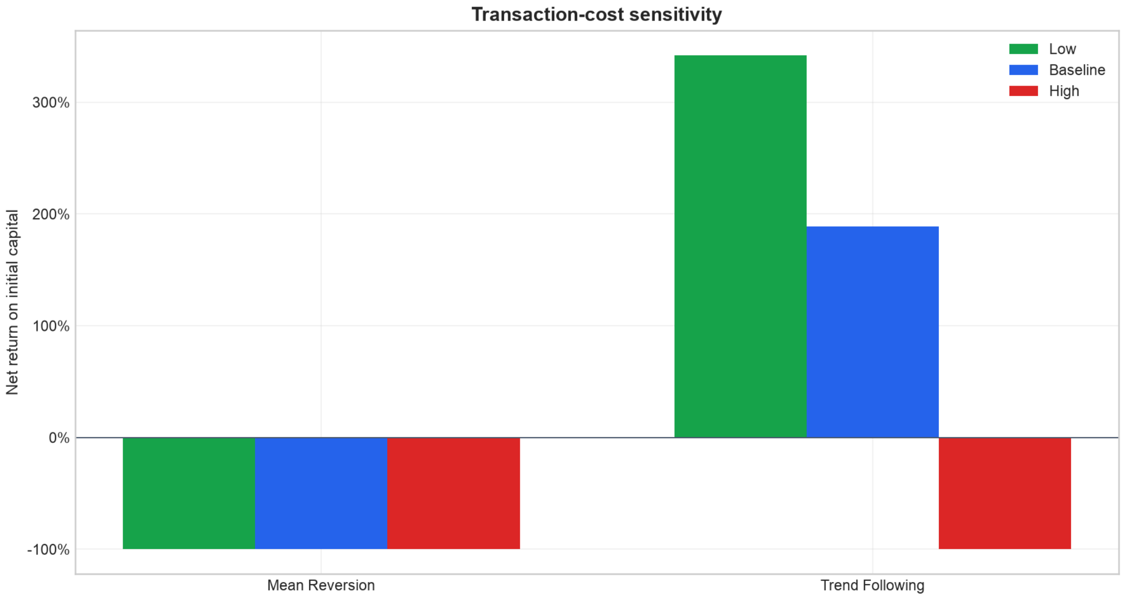

Figure 3. Net return deteriorates sharply as the proportional-cost assumption rises.

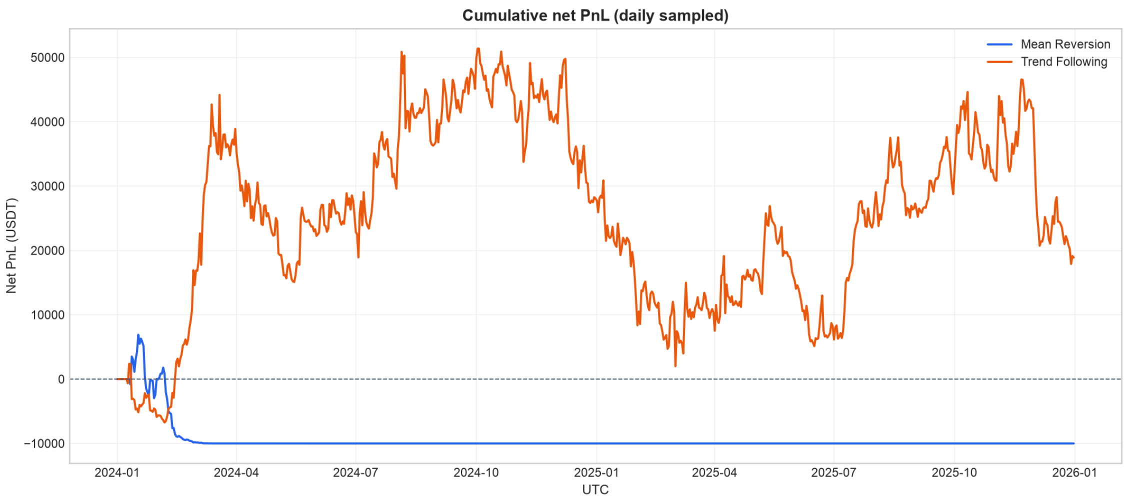

Figure 4. Daily-sampled cumulative net PnL under the baseline cost.

3. Chronological robustness

The same fixed parameters are evaluated separately in 2024 and 2025. No parameter optimization is performed on the holdout. Because the repository was created retrospectively from an earlier full-sample analysis, this split is illustrative rather than a pristine deployment-grade out-of-sample test.

Period	Strategy	Net return	Sharpe	Max drawdown	Final value
Development 2024	Trend following	+280.69%	0.604	-74.84%	38,068.88
Holdout 2025	Trend following	-100.00%	-1.446	-100.00%	0.01

The development period creates the apparent full-sample success. In the 2025 holdout, the same trend rule produces a -1.446 Sharpe ratio and almost total capital loss. This regime instability dominates the investment interpretation.

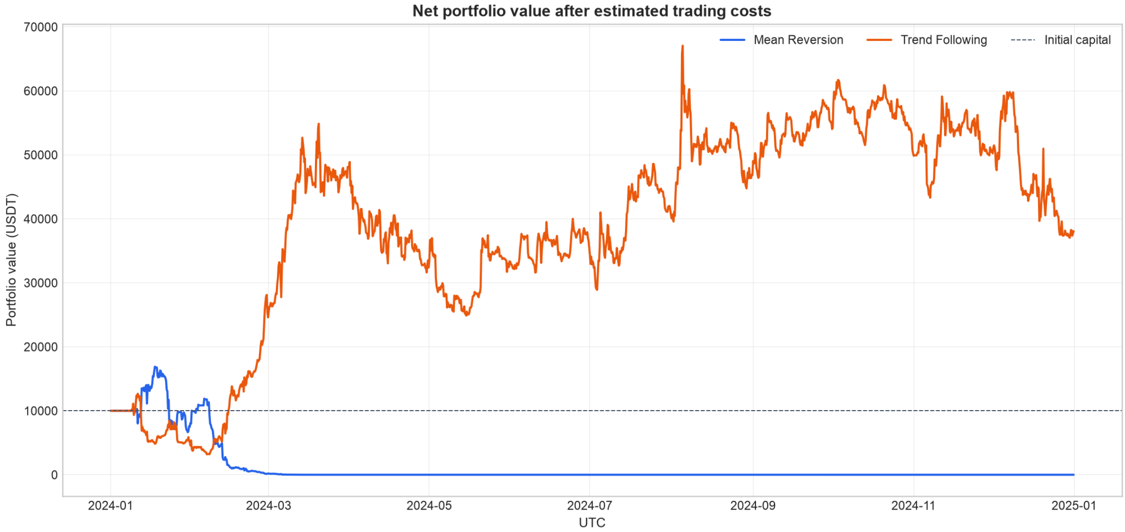

Figure 5. Development-period net equity curves.

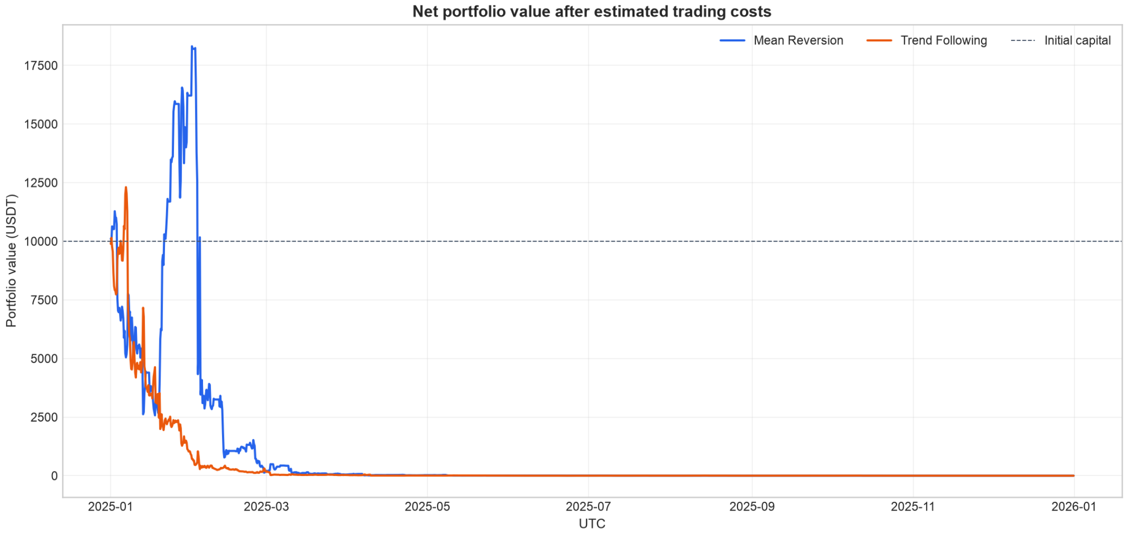

Figure 6. Illustrative holdout net equity curves.

4. Engineering controls

The analysis is implemented as an installable Python package rather than a notebook-only script. Strict YAML configuration defines each experiment. A CLI validates cached data, runs experiments, and returns non-zero codes for invalid inputs. Artifacts are generated in a staging directory and published only after all metrics and figures succeed.

Automated checks

Nineteen tests cover signal lagging, exposure bounds, data repairs, risk-free alignment, cost monotonicity on controlled fixtures, capital floors, hand-checked metrics, CLI behavior, a network-free integration run, and accepted baseline regression values. GitHub Actions runs Ruff and pytest on Python 3.11 and 3.12.

Limitations

The execution model omits explicit exchange fees, dynamic spreads, market impact, funding, latency, partial fills, and survivorship controls. The sample covers only three assets and two years. The outlier rule is a defensive data-engineering demonstration, not a substitute for multi-vendor reconciliation. No claim of statistical significance or future profitability is made.

Next research steps

Useful extensions include volatility-targeted sizing, risk budgets, quote-based spread and impact estimates, walk-forward evaluation, bootstrap uncertainty intervals, a wider survivorship-controlled universe, and a forward paper-trading period.

Reproducibility and sources

The repository documents installation, configuration, CLI reproduction, committed metrics, and public source attribution. Market bars come from Binance Public Data; the risk-free proxy is FRED series DFF. Provider data remains subject to provider terms. The authored code is MIT licensed.

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